

Designing prompts for coding assignments

A good prompt to start building a coding assignment needs to include enough context for the LLM to produce a result that can be refined. Simply asking for code that can produce an outcome leaves too many gaps in the context for the LLM to build. Some LLMs will ask for more information, while others will just build leaving you to do more testing and refinement.

PROVIDE CONTEXT

Example – Target output is a Colab to calculate V_{max} and K_m

Poor prompt: “Generate a Colab for students to calculate V_{max} and K_m from kinetics data”

What’s missing:

- 1) Language
- 2) Packages to use
- 3) What level of student
- 4) Input data format
- 5) Input data file type
- 6) Libraries to use
- 7) Which equation to use to fit
- 8) Clear description of the output and goals
- 9) Other requirements or goals

Better: “I am teaching a lab class of 3rd year Biochemistry majors and need a Python Colab which will take a .csv file with concentration and rate data and fit it to the Michaelis-Menten equation ($rate = (V_{max} * [S]) / (K_m + [S])$) using scipy and pandas. The Colab will guide students through fitting their data to calculate V_{max} and K_m and produce a plot with the fitting line. The data should be fit directly and not use approximations such as Lineweaver-Burk. The plot should be made in matplotlib and I want to show the residuals fit and to generate a separate table of the fitting values with errors.”

Other information you could provide:

- 1) Textbook chapter (link or PDF) to give student background level
- 2) Current assignment worksheet or Excel if you want specific items in the output
- 3) Level of code to show in the Colab (remember you can hide code)

DISCUSS BEFORE BUILDING

You know your student population better than the LLM! Have the LLM show you what will happen at each step so you can provide feedback on the steps and how they are constructed.

Add a phrase like this to your prompt:

“Brainstorm and outline the Colab before writing any code. I want to go block by block through the Colab”

“Which libraries or packages do you recommend to handle this data and produce the outputs? Provide me pros and cons of each package.”

USE A TEMPLATE

Make a template of the Colab that has your desired goals and explanations OR a provide Colab that you want to adapt.

Potential information to include in the template:

- 1) Your library install and import commands
- 2) The assignment guide or text prompts in the Colab to show the flow
- 3) Command to get files from Google Drive/local folder/GitHub repository
- 4) Code framework such as data loading and preview steps or plot generating code

Vocabulary Sheet

A reference for describing what you want, using the right words to get it.

1. General Prompt-Writing Terms

Term	Short Definition
Context	Background info you supply (student level, course goals, data format) that shapes how the LLM builds the assignment.
Cell	One block of code (or text) in a notebook that runs independently.
Library / Package	A collection of pre-written code you import to do a task (e.g., pandas for tables, matplotlib for plots) instead of writing it from scratch.
Import statement	The line of code (e.g., <code>import pandas as pd</code>) that loads a library so you can use it.
Function	A named, reusable block of code that takes inputs and returns an output (e.g., <code>curve_fit()</code>).
Parameter / Argument	A value you pass into a function to control what it does (e.g., telling <code>curve_fit</code> which equation to use).
Variable	A named container that stores a value (a number, a table, a list) so it can be reused later in the code.

2. Data & File Terms

Term	Short Definition
DataFrame	A pandas table (rows and columns) — the standard way Python holds spreadsheet-like data.
CSV file	"Comma-separated values" — a plain-text spreadsheet format; the most common input file type for these assignments.
Column / Field	A single variable in your data table (e.g., "substrate concentration").

Term	Short Definition
Array	A list of numbers (from numpy) that math operations can be applied to all at once — used to hold your x and y data for fitting.
NaN	Not a number or not available number. Typically inserted during data cleaning for missing data or values that don't match the column type (i.e. a word in a column full of numbers)
Missing data	Empty or invalid data points; worth specifying how you want these handled (dropped, flagged, etc.).

3. pandas (data handling)

Term	Short Definition
<code>read_csv()</code>	The pandas function that loads a CSV file into a DataFrame.
Column selection	Pulling out a specific column of data to use (e.g., just the concentration values).
Data cleaning	Removing or fixing bad rows (blanks, typos, outliers) before analysis — say if you want this step included.

4. numpy (numerical operations)

Term	Short Definition
Array operations	Applying a calculation to every value in a dataset at once (e.g., converting all concentrations from mM to μM).
<code>linspace</code> / <code>arange</code>	numpy functions that generate an evenly spaced range of numbers — often used to draw a smooth fitted curve.

5. matplotlib (plotting)

Term	Short Definition
Plot / Axes	The chart itself and the x/y coordinate system it's drawn on.
Scatter plot	A plot showing individual data points (e.g., raw rate vs. concentration measurements).
Line plot / Fit line	A smooth curve drawn through or over the data — typically the model's best-fit prediction.
Legend	The key on a plot identifying what each color/symbol represents.
Axis labels / Title	Text identifying what's plotted and in what units — worth specifying explicitly (LLMs won't guess your units).
Residuals plot	A secondary plot showing the difference between each data point and the fitted curve — helps students judge fit quality.

6. scipy.optimize.curve_fit (nonlinear regression)

Term	Short Definition
Model equation	The mathematical formula you're fitting to (e.g., Michaelis-Menten: $\text{rate} = V_{\text{max}} \cdot [\text{S}] / (K_{\text{m}} + [\text{S}])$) — spell this out explicitly in your prompt.
Fitting parameters	The unknown values the fit solves for (e.g., V_{max} and K_{m}) — as opposed to your x/y data, which is known.
Initial guess (p0)	Starting values for the parameters that curve_fit uses before refining them — sometimes needed for the fit to converge correctly.
Convergence	When the fitting algorithm successfully settles on a solution rather than failing or looping indefinitely.
Covariance matrix / Standard error	Output from curve_fit describing how uncertain each fitted parameter is — needed if you want error bars or a table of values $\pm$ error.

Term	Short Definition
Linearization (e.g., Lineweaver-Burk)	An older approach that transforms a curved relationship into a straight line to fit it — worth explicitly ruling in or out, since it changes both the math and the error behavior.
Goodness of fit (R^2 , residuals)	Metrics/plots that show how well the model matches the data — specify if you want these reported.

7. RDKit (cheminformatics)

Term	Short Definition
RDKit	A library for working with chemical structures in Python — reading molecules, calculating properties, and drawing structures.
SMILES	A text string that encodes a molecule's structure (e.g., <chem>CC(=O)Oc1ccccc1C(=O)O</chem> for aspirin) — the most common way to hand a molecule to RDKit.
Mol object	RDKit's internal representation of a molecule, created from a SMILES string — most RDKit functions expect this, not the raw text.
Descriptor	A calculated numeric property of a molecule (molecular weight, LogP, number of hydrogen bond donors, etc.).
Fingerprint	A compact numeric representation of a molecule's structure, used to compare or search for similar molecules.
Substructure search	Checking whether a molecule contains a particular chemical pattern or functional group.
Tanimoto similarity	A standard score (0–1) for how structurally similar two molecules are, usually computed from their fingerprints.

7. Common Functions by Package

Useful for naming specifically what you want the code to do, rather than describing it in a roundabout way.

pandas

Function	What it does
<code>pd.read_csv()</code>	Load a CSV file into a DataFrame
<code>df.head()</code>	Preview the first few rows of a table
<code>df.dropna()</code>	Remove rows with missing data
<code>df['column']</code>	Select a single column
<code>df.describe()</code>	Quick summary stats (mean, min, max, etc.) for each column
<code>df.groupby()</code>	Group rows by a category and summarize each group
<code>df.to_csv()</code>	Save a DataFrame back out to a CSV file

numpy

Function	What it does
<code>np.array()</code>	Convert a list of numbers into a numpy array
<code>np.linspace()</code>	Generate evenly spaced numbers over a range (e.g., for a smooth fit line)
<code>np.mean()</code> / <code>np.std()</code>	Calculate the average / standard deviation of a dataset
<code>np.sqrt()</code>	Square root — often used when computing errors from a covariance matrix
<code>np.log()</code> / <code>np.exp()</code>	Natural log / exponential — common in kinetics and binding equations
<code>np.polyfit()</code>	Fit data to a simple polynomial (e.g., a straight line) — a simpler alternative to <code>curve_fit</code>

matplotlib

Function	What it does
<code>plt.plot()</code>	Draw a line plot
<code>plt.scatter()</code>	Draw a scatter plot of individual data points
<code>plt.xlabel()</code> / <code>plt.ylabel()</code>	Label the x-axis / y-axis
<code>plt.title()</code>	Add a title to the plot
<code>plt.legend()</code>	Show a legend identifying each data series
<code>plt.savefig()</code>	Save the plot as an image file
<code>plt.subplots()</code>	Create a figure with multiple plots (e.g., data + residuals side by side)

scipy (scipy.optimize)

Function	What it does
<code>curve_fit()</code>	Fit a model equation to data and return the best-fit parameters
<code>np.diag(pcov)</code>	Pull the variances off the covariance matrix returned by <code>curve_fit</code> (paired with <code>np.sqrt()</code> for standard errors)
<code>minimize()</code>	General-purpose optimization — useful if you need something beyond curve fitting
<code>curve_fit(..., p0=...)</code>	Passing an initial guess for the parameters to help the fit converge
<code>curve_fit(..., bounds=...)</code>	Restrict fitted parameters to a realistic range (e.g., Km can't be negative)

RDKit

Function	What it does
Chem.MolFromSmiles()	Convert a SMILES string into a Mol object RDKit can work with
Chem.MolToSmiles()	Convert a Mol object back into a SMILES string
Descriptors.MolWt()	Calculate a molecule's molecular weight
Descriptors.MolLogP()	Estimate a molecule's LogP (a measure of lipophilicity)
Draw.MolToImage()	Generate a 2D image of a molecule's structure
AllChem.GetMorganFingerprint()	Compute a fingerprint used for similarity comparisons
DataStructs.TanimotoSimilarity()	Calculate the similarity score between two fingerprints

9. Tips for Using This Sheet

- You don't need to use every term — but naming the library, the model equation, and what output you expect (plot type, table, error values) turns a vague prompt into a buildable one.
- If a term feels shaky, it's fine to just describe what you want in plain language — "show me how confident the fit is" works even if you don't say "standard error."
- Pair this sheet with the "Better" prompt example in the main handout as a template to fill in for your own assignment.